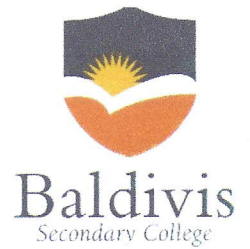


Year 10 - 2016

Investigation 2 Validation

Semester One Term 2

Probability



This investigation will be completed in class under test conditions to fully test your understanding of probability. No guidance or hints will be provided for this. You may use the in class section and any other A4 note page to in the validation to help you.

Calculators are allowed.

Time allocation 50 minutes.

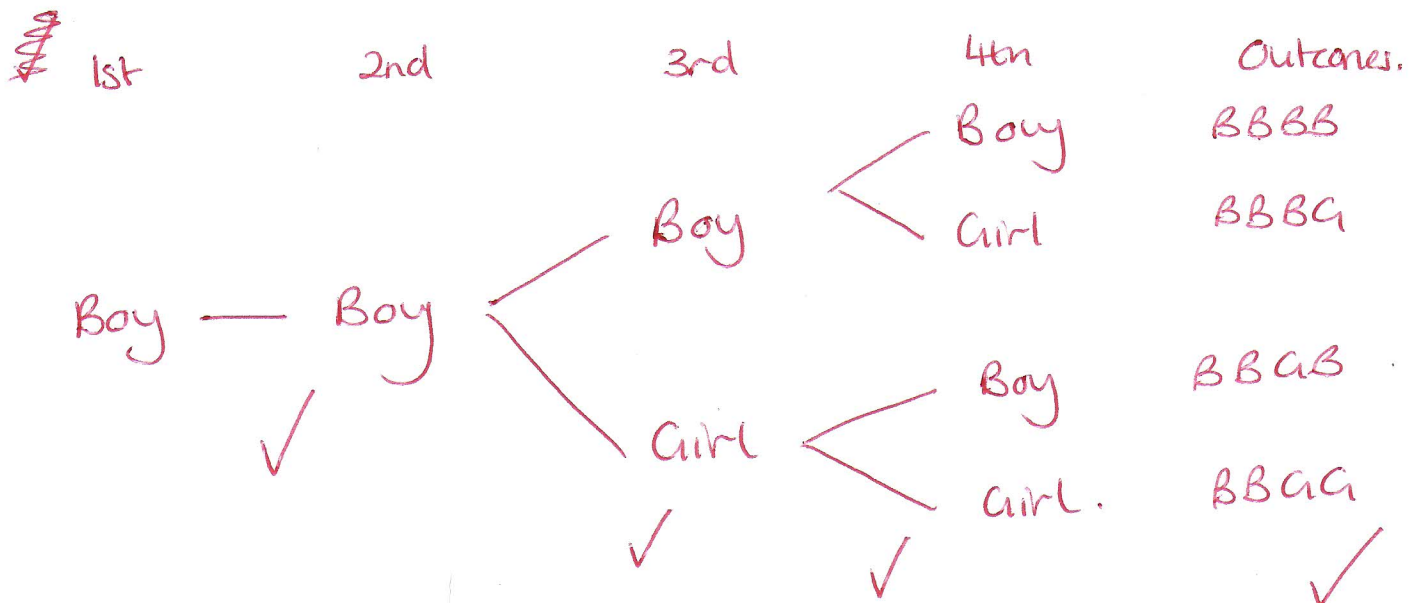
AIM: In this assessment you will be investigating the results of two and three step chance experiments, assigning probabilities to outcomes and determining the probabilities of events.

Name: Mark Incey Class: _____ Mark: _____ /22

1. [4, 2, 2, 2 = 10]

The Lee's have four children consisting of two boys followed by a set of twins.

a) Draw a tree diagram to show the different combination of sexes of the children.



- b) Assume that each outcome in your tree diagram is equally likely.
- Find the probability that all four children are boys.

$$\frac{1}{4}$$



- Find the probability that the twins are both girls.

$$\frac{1}{4}$$



- Given the fourth child is a boy, find the probability the twins are of different sexes.

$$\frac{1}{2}$$



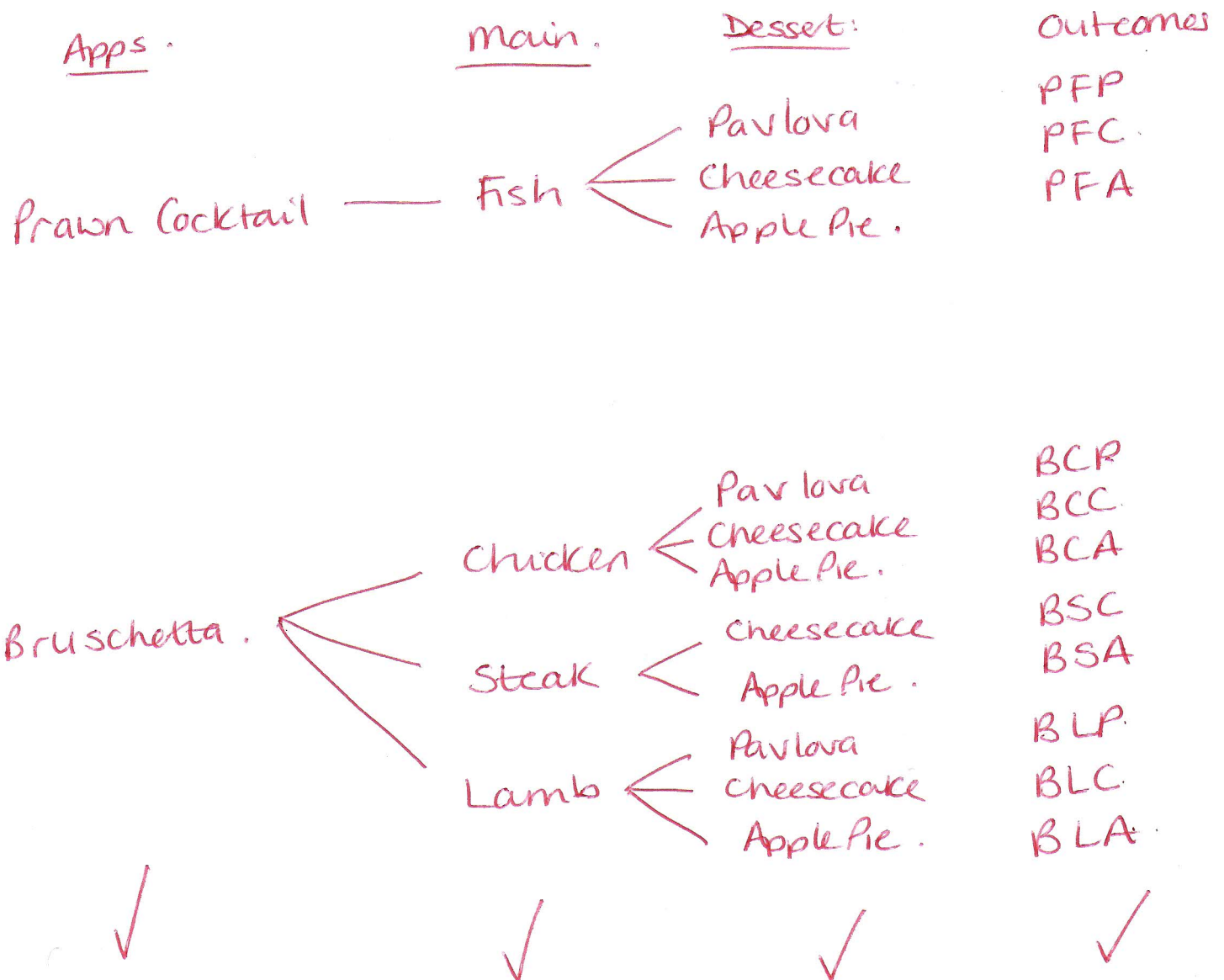
2. $[4, 2, 2, 2, 2 = 12]$

Allies' Café offers the following menu:

Appetisers: Prawn Cocktail or Bruschetta
Main Meal: Fish or Chicken or Steak or Lamb
Dessert: Pavlova or Cheesecake or Apple pie

Jenny orders an appetiser, one item for the main meal and one dessert. Jenny will not have Steak with Pavlova and must have Prawn Cocktail with Fish and Fish only with Prawn cocktail.

a) Draw a clearly labelled tree diagram to show Jenny's possible appetiser/ main meal/ dessert combinations.



b) If each of Jenny's choices was equally likely, find the probability that Jenny:

i. Chose Chicken as a main meal

$$\frac{3}{11}$$



ii. Did not choose Cheesecake as a dessert

$$\frac{7}{11}$$



iii. Chose Cheesecake given she chose Chicken

$$\frac{1}{3}$$



iv. Chose Chicken given she chose Cheesecake

$$\frac{1}{4}$$



END OF INVESTIGATION